

# How Helpful Is **BLOCKCHAIN** TECHNOLOGY In Cloud Storage



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*Blockchain has revolutionised the Internet. This new and ingenious technology allows blocks of data to be distributed across ledgers, without any central administering authority, but with the essential requirement that the data be validated by participants. This makes the data flexible and secure. Blockchain technology can be used to great advantage in cloud storage solutions*

**B**lockchain technology is one of the greatest inventions of today. You would have heard about digital currency, cryptocurrency or Bitcoin. These exist because of blockchain technology. In this article, we will first understand what blockchain technology is and how it works. Then we will look at how we can correlate cloud computing with blockchain technology.

Blockchain was invented in 2008 by a person or a group using the name Satoshi Nakamoto. Till date, no one knows the real identity of Satoshi Nakamoto. Blockchain has evolved into something very powerful since then.

Blockchain technology is built using peer-to-peer networking. Anyone can join the network and there is no central authority to manage it. It is operated by people, called miners, who lend their computing power to the network to solve complex algorithms. Machines run by miners perform the necessary functions to solve the algorithms, and for that they are rewarded with a fee (for lending their computing power). In the case of Bitcoin, that complex

"The blockchain is an incorruptible digital ledger of economic transactions that can be programmed to record not just financial transactions but virtually everything of value."

—Don and Alex Tapscott,  
authors of 'The Blockchain Revolution'

algorithm is SHA-256.

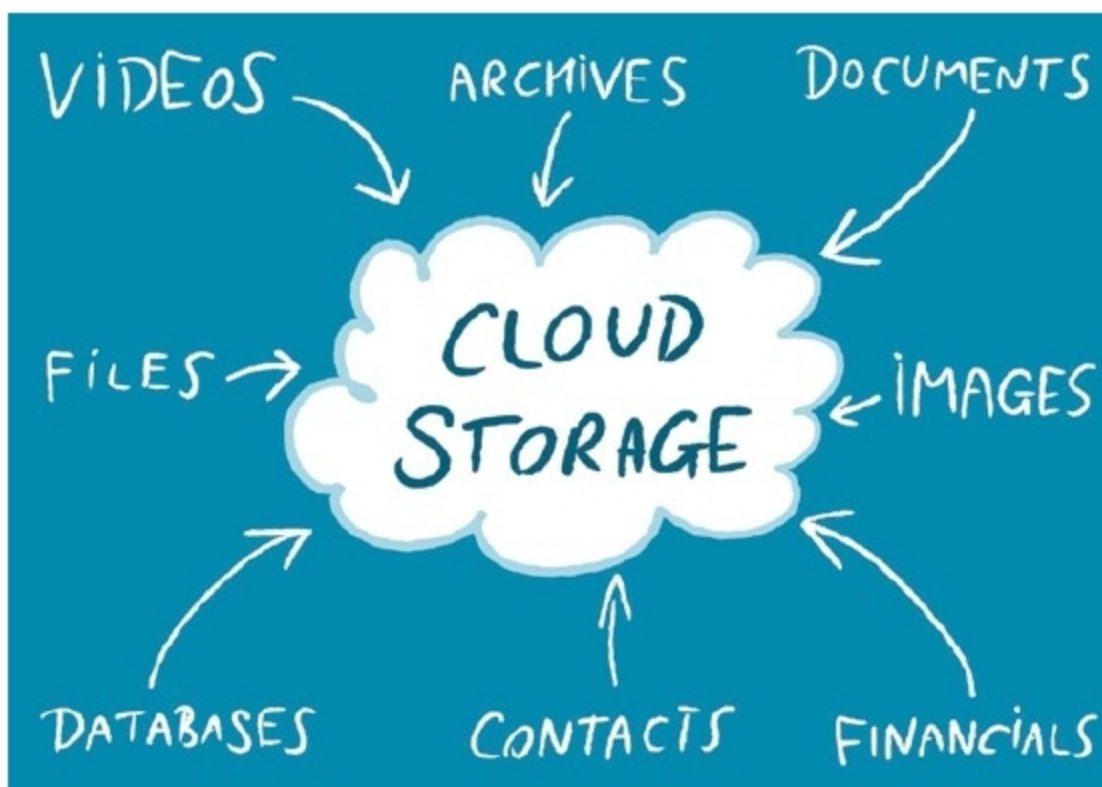
In a blockchain, each block stores data of the transaction, its hash code and the previous block's hash code. So whenever a new block is created, it is validated by a majority of peers or miners on that network. If anyone tries to change the data in one block, the entire blockchain becomes invalidated. So it is nearly impossible for an individual to change any of the attached blocks. This is how a blockchain remains secure and managed. Also, all data are encrypted for security purposes.

Bitcoin works on the same technology. It is a virtual currency that is mined using blockchain technique. The value of one Bitcoin is approximately ₹ 469,636, at the time of printing. As you can imagine, it is virtually like gold.

Blockchain technology is not just used for cryptocurrencies but can be used in many other sectors also. Potential use cases are video or audio streaming, cloud storage and much more. Today, a lot of research is going on to study how to tap into the power of this technology and apply it across various verticals.

## Cloud storage

Google Drive, Dropbox and Gmail are basic examples of cloud storage. Cloud storage allows users to store data online and, hence, access it anywhere and at any time—that is the biggest advantage. Nowadays, we store most of our data in digital format, having moved beyond the stage of managing bundles of physical files. Government agencies are also leveraging the advantages of cloud storage



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